

ARUNIMA SURENDRAN

Applied AI Engineer | Technical Writer



Physics graduate with a background in applied AI engineering and technical writing, currently at Intel's OpenEdge team. I build proof-of-concept video analytics pipelines using OpenVINO, DLStreamer, and Intel Geti, and write developer-focused blogs and documentation for edge AI products. Previously worked in cloud-based environments building RAG pipelines, vision models, and OCR systems. I bring engineering depth to content and content clarity to engineering work.

Bachelor of Science - Physics

2017 - 2020

Central University of Pondicherry

CGPA: 7.67/10

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WORK EXPERIENCE

Developer: Technical Marketing III | Nov, 2025 - May, 2026

Intel Corporation (External Contractor via Harvey Nash Inc)

- Built proof-of-concept video analytics pipelines and use-case demos for the OpenEdge Platform and OpenVINO toolkit, covering worker safety, crowd density, and anomaly detection scenarios.
- Authored developer-focused and high-level technical blogs for Edge AI Suite releases, including the Open Edge Platform 2026.0 launch.
- Contributed to and reviewed technical documentation for the OpenEdge Platform, including gap-filling PRs to improve developer onboarding.
- Conducted pre-release testing of edge AI applications, identifying and reporting bugs ahead of product launches.
- Participated in the product marketing messaging process as a subject matter resource for edge AI capabilities.

Technical Writer/ ML Engineer | June, 2025 - September, 2025

E2E Networks, Bengaluru, Karnataka

- Designed, built, and fine-tuned RAG, Instruction-tuned, Vision, and OCR models, and MCP Servers demonstrating end-to-end expertise in model development.
- Authored and published technical blogs that consistently ranked in Google's Top 3 search results and secured brand visibility in leading generative AI platforms through Generative Engine Optimization (GEO).
- Built Google Drive MCP servers using distinct methods: OAuth Client, GDrive API, Rclone mount, and Cookie-based access to enable flexible data integration.

Content Writer | May, 2024 - May, 2025

Qicpic Innovations Pvt. Ltd., Bengaluru, Karnataka

- Developed and trained text-to-text LLMs to automate optimized marketing content.
- Created website/UI/UX content and automated content generation.
- Produced SEO-friendly content for digital marketing, including blogs, video scripts, and emails.

Freelance Content Writer | Oct, 2020 - Apr, 2024

Creative Writing: Created B2B/B2C social media content, blogs, emails, and SEO-optimized website copy while developing content strategies, marketing plans, and editorial calendars.

Academic Writing: Simplified complex Physics and Mathematics concepts for students with clear explanations and solutions.

Subject Matter: Solved Physics and Mathematics problems with detailed explanations and concept clarifications.

FEATURED OPEN-SOURCE CONTRIBUTIONS

- Worker Safety and Fall Detection Edge Application: Multimodel edge AI pipeline for real-time worker fall detection using Intel Geti and OpenVINO IR, combining a bounding box ratio heuristic, temporal aggregation, and keypoint detection as a multi-stage filtering approach to balance accuracy and compute efficiency.
- Crowd Density Estimation Model: Developed a real-time crowd density estimation pipeline using CSRNet with a VGG-16 frontend and dilated convolution backend, converted directly to OpenVINO IR via `ov.convert_model()` to avoid ONNX intermediaries and operator compatibility issues.
- Driver Fatigue Detection Model: Built a driver fatigue detection system using multi-model cascades and temporal aggregation, applying behavior-level reasoning across facial and eye landmark signals for robust real-time inference on edge hardware.
- Fine-tuned the gpt-oss-20b model for medical reasoning using an SFT dataset of ~90k rows and an evaluation dataset of 1.7M rows across 27 subsets, achieving a BERTScore of 0.834. Released a comprehensive [Hugging Face model card](#) detailing methodology, evaluation, and ethical considerations.
- Anomaly Detection Model: Trained a convolutional autoencoder on normal casting images for unsupervised defect detection, using reconstruction error thresholding to flag anomalies. Exported to ONNX and OpenVINO IR for edge deployment on Intel hardware.

FEATURED ARTICLES

- [Real-Time Crowd Density Estimation Using CSRNet and OpenVINO](#)
- [Advancing Edge Intelligence: What's New in Open Edge Platform 2026.0](#)
- [A Developer's Guide to Edge AI Workloads with Metro AI Suite](#)
- [Building a Domain-Specific Medical Reasoning Model with Supervised Fine-Tuning on GPT-OSS-20B](#)

SKILLS & TOOLS

- Technical Writing: XML, Markdown, LaTeX, Swagger, Postman, MadCap Flare, SEO & GEO
- Programming & Frameworks: Python, Node.js, HTML, FastAPI, FastMCP, PyTorch
- ML & NLP: Hugging Face (Transformers, Diffusers), Transfer Learning, Fine-tuning, Sentiment Analysis, Zero-shot Classification, NER, LLMs, Vision & Speech Models, Ollama, RAG Pipelines, OCR
- Computer Vision & Edge AI: OpenVINO, Intel Geti, DLStreamer, OpenVINO IR, Model Conversion & Optimization, ONNX, CSRNet, YOLOX, Anomalib
- APIs & Workflow Automation: REST APIs, Hugging Face Hub, n8n, Langflow, LangGraph, MCP (Streamable, Stdio, FastMCP)
- Deployment & Tools: Docker, Kubernetes, Streamlit, Flask, Gradio, Cloudflare Pages, iNaturalist (field identification)

LANGUAGES

- Native level of fluency in English: Speaking, Writing and Translating
- Native speaker of Malayalam
- Basic knowledge in French and Spanish

REFERENCES

Dr. Ashutosh Kumar,
Technical Marketing Engineer,
Intel Corporation